Nakuja Project

Solid Propulsion Team

One-Grain Static Test — Control Grain Repeat

Document Version:	1.0 (Post-Execution Report)
Motor Configuration:	N4, 1-Grain, Aluminium Casing, BATES Geometry (Control)
Date of Execution:	Thursday, 30 July 2026
Location:	Baseball Pitch (consistent with prior test venue)
Prepared by:	Paschal Peters (Data Recorder / Timer)
Outcome:	**SUCCESS** — Performed as expected, no issues

1 Purpose

This report documents a repeat static test of the **control grain** (baseline formulation, no surfactant or suppressant additives) conducted on 30 July 2026, at the same venue and under materially the same conditions as prior tests. This test serves as a fresh baseline data point against which the team's various additive trials — the 65:35 and 60:40 KNO_3:Sorbitol tests, the 0.1% SLS surfactant trial, and the 1% sodium bicarbonate suppressant trial — can be more confidently compared, since it re-establishes how the unmodified grain performs under current test conditions.

2 Test Overview

Motor	N4, 1-Grain, Aluminium Casing (Control formulation)
Burn Time	≈2.51 s
Faults or Anomalies	None
Outcome	**Success** — performed as expected

3 Theoretical vs. Experimental Comparison

Parameter	Experimental	Simulated
Total Impulse	968 Ns	1014 Ns
Peak Thrust	≈718 N	≈378 N
Burn Time	2.51 s	2.85 s

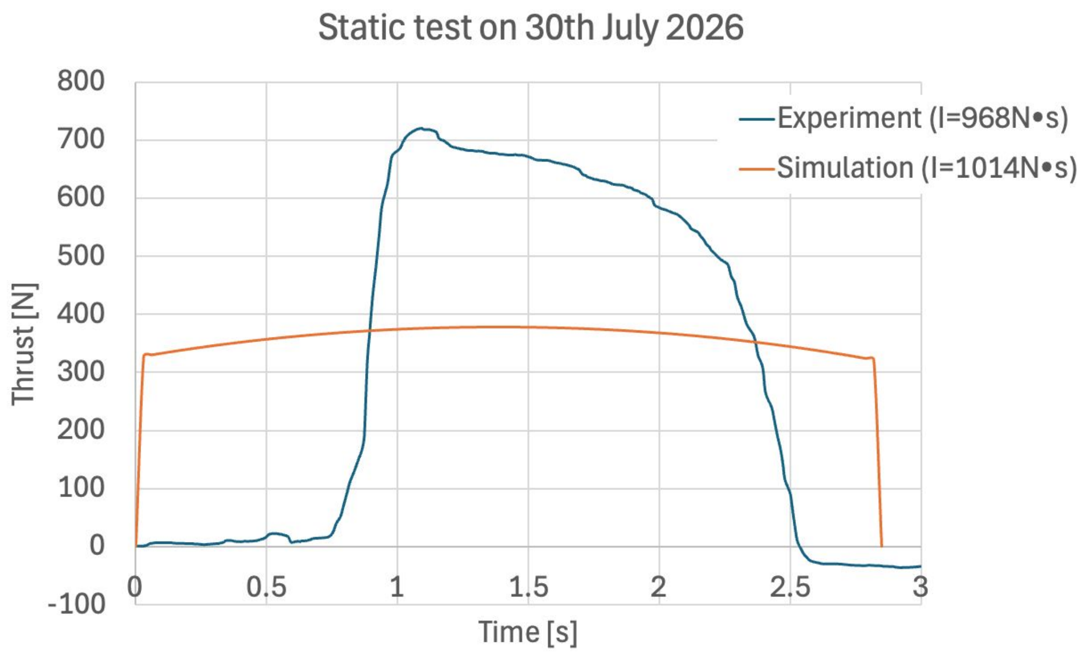

Figure 1: Thrust curve comparison for the control grain static test, 30 July 2026: experimental (blue) vs. simulated (orange).

Relative to the simulation, total impulse was within approximately **4.5%** of the predicted value, and burn time was within approximately **12%** — both a substantially closer match to theory than the earlier 65:35 and 60:40 grain tests, which ran 3–4 times longer than predicted. Peak thrust, by contrast, nearly **doubled** the simulated value (≈718 N vs. ≈378 N), while occurring at a similar point in the burn (≈1.1 s experimental vs. ≈1.3 s simulated).

This combination — a burn time and impulse close to theoretical prediction, alongside a sharply higher peak thrust — is consistent with the grain regressing at closer to its intended (design) burn rate overall, but with a more aggressive pressure build-up during the early-to-mid burn than the ideal BATES model assumes, before tapering off in a longer tail than the simulation's sharp cutoff. This is a marked improvement in burn-time fidelity over the earlier 65:35/60:40 grain tests, even though the peak thrust deviation itself is now the more notable gap from theory.

4 Outcome Classification

This test is classified as a **SUCCESS**. The control grain performed as expected, with a burn time of ≈2.51 s — close to the theoretical prediction of 2.85 s — and no faults, anomalies, or issues reported during the test.

5 Recommendations and Next Steps

- Retain this test as the current reference control dataset for comparison against ongoing and future additive trials (surfactant, suppressant, or ratio-modified grains).
- Investigate the cause of the elevated experimental peak thrust relative to simulation (≈718 N vs.

$\approx$378 N), as this is now the primary remaining discrepancy between theory and experiment for the control formulation.

- Continue tracking burn-time fidelity across future control repeats to confirm that the improved agreement seen here (2.51 s vs. 2.85 s) is consistent and repeatable, rather than a single favourable result.
- Use this dataset alongside the 1% sodium bicarbonate suppressant trial (once repeated with working instrumentation) to assess how much the suppressant shifts peak thrust and burn time relative to this fresh control baseline.

Nakuja Project — Solid Propulsion Team, KNSB Solid Rocket Motor Development